

ASSIGNMENT 4_GROUP(41).

1. Determine the observability matrix W_o of the driveline system! Is the system observable?

Problem 1:- The state-space representation of driveline

$$\dot{x} = Ax + Bu + Dv$$

$$y = Cx + w.$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \Rightarrow \begin{aligned} x_1 &= \text{engine speed} \\ x_2 &= \text{wheel speed} \\ x_3 &= \text{driveshaft torsion.} \end{aligned}$$

The model of the driveline system equation is

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}} & \frac{-c_s}{J_f r_{gear}} \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{c_s}{J_c} \\ \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \frac{1}{J_f} \\ 0 \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ \frac{1}{J_c} \\ 0 \end{bmatrix} v$$

$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + w$$

$$\text{where } A = \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}} & \frac{-c_s}{J_f r_{gear}} \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{c_s}{J_c} \\ \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix} \quad B = \begin{bmatrix} \frac{1}{J_f} \\ 0 \\ 0 \end{bmatrix} \quad D = \begin{bmatrix} 0 \\ \frac{1}{J_c} \\ 0 \end{bmatrix}$$

$$C = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

observability matrix W_o

$$W_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

Matrix CA

$$CA = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}} & \frac{-C_s}{J_f r_{gear}} \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{C_s}{J_c} \\ \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{C_s}{J_c} \end{bmatrix}$$

Matrix CA²

$$CA^2 = \begin{bmatrix} \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{C_s}{J_c} \end{bmatrix} \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}} & \frac{-C_s}{J_f r_{gear}} \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{C_s}{J_c} \\ \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix}$$

$$CA^2 = \begin{bmatrix} \frac{-d_s^2}{J_c J_f r_{gear}^3} - \frac{d_s^2}{J_c^2 r_{gear}} + \frac{C_s}{J_c r_{gear}} & \frac{d_s^2}{J_c J_f r_{gear}^2} + \frac{d_s^2}{J_c^2} - \frac{C_s}{J_c} & \frac{-d_s C_s}{J_c J_f r_{gear}^2} \\ \frac{-d_s C_s}{J_c^2} & +0 & \end{bmatrix}$$

W₀ =

$$\begin{bmatrix} 0 & 1 & 0 \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{C_s}{J_c} \\ \frac{-d_s^2}{J_c J_f r_{gear}^3} - \frac{d_s^2}{J_c^2 r_{gear}} + \frac{C_s}{J_c r_{gear}} & \frac{d_s^2}{J_c J_f r_{gear}^2} + \frac{d_s^2}{J_c^2} - \frac{C_s}{J_c} & \frac{-C_s d_s}{J_c J_f r_{gear}^2} - \frac{C_s d_s}{J_c^2} \end{bmatrix}$$

W₀ is equal to number of states, the system is observable
The rank of observability matrix is 3.

The W₀ matrix is observable.

2. Is the system observable from the engine speed or the driveshaft torsion?

Problem 2 :- To check observability from Engine speed. ($n, = 1$)

The output matrix will be $C = [1 \ 0 \ 0]$

$$W_E = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

The matrix CA :

$$CA = [1 \ 0 \ 0] \begin{bmatrix} \frac{-d_s}{J_f r_{gear}} & \frac{d_s}{J_f r_{gear}} & \frac{-c_s}{J_f r_{gear}} \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{c_s}{J_c} \\ \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}} & \frac{-c_s}{J_f r_{gear}} \end{bmatrix}$$

The matrix CA^2 :

$$CA^2 = \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}} & \frac{-c_s}{J_f r_{gear}} \end{bmatrix} \begin{bmatrix} \frac{-d_s}{J_f r_{gear}} & \frac{d_s}{J_f r_{gear}} & \frac{-c_s}{J_f r_{gear}} \\ \frac{d_s}{J_c r_{gear}} & \frac{-d_s}{J_c} & \frac{c_s}{J_c} \\ \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix}$$

$$CA^2 = \begin{bmatrix} \frac{+d_s^2}{J_f^2 r_{gear}^3} + \frac{d_s^2}{J_c J_f r_{gear}^2} & \frac{-c_s}{J_f r_{gear}} & \frac{-d_s^2}{J_f^2 r_{gear}^3} - \frac{d_s^2}{J_f J_c r_{gear}^2} + \frac{c_s}{J_f r_{gear}} & \frac{c_s d_s}{J_f^2 r_{gear}^3} \\ \frac{c_s d_s}{J_c J_f r_{gear}} & 0 \end{bmatrix}$$

$$W_E =$$

$$W_E = \begin{bmatrix} 1 & 0 & 0 \\ \frac{-d_s}{J_f r_{gear}^2} & \frac{d_s}{J_f r_{gear}^2} & \frac{-c_s}{J_f r_{gear}^2} \\ \frac{d_s^2}{J_f^2 r_{gear}^4} + \frac{d_s^2}{J_c r_{gear}^2} - \frac{c_s}{J_f r_{gear}^2} & \frac{-d_s^2}{J_f^2 r_{gear}^4} - \frac{d_s^2}{J_c r_{gear}^2} + \frac{c_s}{J_f r_{gear}^2} & \frac{+c_s d_s}{J_f^2 r_{gear}^4} + \frac{c_s d_s}{J_c r_{gear}^2} \end{bmatrix}$$

[w_E is observable]

To check observability from Drive shaft Torsion ($x_3 = 1$).

The output matrix will be $c = [0 \ 0 \ 1]$.

$$W_O = \begin{bmatrix} c \\ cA \\ cA^2 \end{bmatrix}$$

The matrix cA will be

$$cA = \begin{bmatrix} \frac{1}{r_{gear}} & -1 & 0 \end{bmatrix}$$

The matrix cA^2 :

$$cA^2 = \begin{bmatrix} \frac{-d_s}{J_f r_{gear}^3} & \frac{-d_s}{J_c r_{gear}} + 0 & \frac{d_s}{J_f r_{gear}^2} + \frac{d_s}{J_c} + 0 & \frac{-c_s}{J_f r_{gear}^2} - \frac{c_s}{J_c} & 0 \end{bmatrix}$$

$$W_O = \begin{bmatrix} 0 & 0 & 1 \\ \frac{1}{r_{gear}} & -1 & 0 \\ \frac{-d_s}{J_f r_{gear}^3} & \frac{-d_s}{J_c r_{gear}} & \frac{d_s}{J_f r_{gear}^2} + \frac{d_s}{J_c} & \frac{-c_s}{J_f r_{gear}^2} - \frac{c_s}{J_c} \end{bmatrix}$$

$$\det(W_O) = 0 \begin{bmatrix} -1 & 0 & 0 \\ \frac{d_s}{J_f r_{gear}^2} + \frac{d_s}{J_c} & \frac{-c_s}{J_f r_{gear}^2} - \frac{c_s}{J_c} \end{bmatrix} - 0 \begin{bmatrix} \frac{1}{r_{gear}} & 0 \\ \frac{-d_s}{J_f r_{gear}^3} & \frac{-c_s}{J_f r_{gear}^2} - \frac{c_s}{J_c} \end{bmatrix} +$$

$$0 \begin{bmatrix} \frac{1}{r_{gear}} & -1 \\ \frac{-d_s}{J_f r_{gear}^3} & \frac{d_s}{J_f r_{gear}^2} + \frac{d_s}{J_c} \end{bmatrix} = 0 - 0 + 0 \Rightarrow 0$$

W_O Not observable

The system is observable is from Engine speed.

The system is not observable for driveshaft torsion. (In matlab the W_o_tor is showing with rank has 2, please neglect the matlab answer. As hand calculation is proved above where $\det(W_d)$ the system is Not observable. (As discussed in the consultation hour on Friday I.e 11.10.2024, hand calculation is attached).

3. How will the output matrix C change if the driveshaft torque is the output signal? Is the system observable from the driveshaft torque?

Problem 3: The common relation for $T_{\text{shaft}} = c_s \phi + d_s (\omega_{\text{dr}} - \omega_{\text{wr}})$

where: $T_{\text{shaft}} = c_s \cdot \phi + d_s \left(\frac{\omega_{\text{eng}}}{i} - \omega_w \right)$

ω_{eng} = engine speed

c_s = driveshaft's spring w-coefficient

ω_w = wheel speed

d_s = driveshaft's damping w-coefficient

$T_{\text{shaft}} = c_s \cdot \phi + d_s \frac{\omega_{\text{eng}}}{i} - d_s \omega_w$

$y = Cx$ $i = r_{\text{gear}}$

$$C = \begin{bmatrix} \frac{d_s}{r_{\text{gear}}} & -d_s & c_s \end{bmatrix}$$

The observability Matrix $W_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$

$$CA = \begin{bmatrix} \frac{d_s}{r_{\text{gear}}} & -d_s & c_s \end{bmatrix} \begin{bmatrix} -\frac{d_s}{J_f r_{\text{gear}}} & \frac{d_s}{J_f r_{\text{gear}}} & -\frac{c_s}{J_f r_{\text{gear}}} \\ \frac{d_s}{J_c r_{\text{gear}}} & -\frac{d_s}{J_c} & \frac{c_s}{J_c} \\ 1 & -1 & 0 \end{bmatrix}$$

$$CA = \begin{bmatrix} -\frac{d_s^2}{J_f r_{\text{gear}}} - \frac{d_s^2}{J_c r_{\text{gear}}} + \frac{c_s}{r_{\text{gear}}} & \frac{d_s^2}{J_f r_{\text{gear}}} + \frac{d_s^2}{J_c} - c_s & \frac{-c_s d_s}{J_f r_{\text{gear}}} - \frac{c_s d_s}{J_c} + 0 \end{bmatrix}$$

$$CA^2 = \begin{bmatrix} \frac{d_s^3}{J_f^2 r_{\text{gear}}} - \frac{d_s^3}{J_c^2 r_{\text{gear}}} + \frac{c_s}{r_{\text{gear}}} & \frac{d_s^3}{J_f^2 r_{\text{gear}}} - \frac{d_s^3}{J_c^2} + c_s & \frac{c_s^2 d_s}{J_f^2 r_{\text{gear}}} - \frac{c_s^2 d_s}{J_c^2} + 0 \end{bmatrix}$$

$W_{\text{shaft torque}} =$

$$\begin{bmatrix} -\frac{d_s}{r_{\text{gear}}} & -d_s & c_s \\ \frac{-d_s^2}{J_f r_{\text{gear}}} - \frac{d_s^2}{J_c r_{\text{gear}}} + \frac{c_s}{r_{\text{gear}}} & \frac{d_s^2}{J_f r_{\text{gear}}} + \frac{d_s^2}{J_c} - c_s & \frac{-c_s d_s}{J_f r_{\text{gear}}} - \frac{c_s d_s}{J_c} + 0 \\ \frac{d_s^3}{J_f^2 r_{\text{gear}}} - \frac{d_s^3}{J_c^2 r_{\text{gear}}} + \frac{c_s}{r_{\text{gear}}} & \frac{d_s^3}{J_f^2 r_{\text{gear}}} - \frac{d_s^3}{J_c^2} + c_s & \frac{c_s^2 d_s}{J_f^2 r_{\text{gear}}} - \frac{c_s^2 d_s}{J_c^2} + 0 \end{bmatrix}$$

The system is Not observable.

4. Where should the poles be placed?

Problem 4: The characteristic polynomial equation.

$$s^2 + 2\zeta\omega_n s + \omega_n^2, \quad \omega_n = 5, \quad \zeta = 1$$

1) The poles for the feedback controller are

$$s_1, s_2 = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$= -5 \pm j0$$

feedback poles = $s_1 = -5, \quad s_2 = -5, \quad s_3 = -3\zeta\omega_n = -3 \times 1 \times 5$

$$s_3 = -15$$

s_1 and s_2 are the dominant poles.
 s_3 is the faster pole.

2) The observer poles are:-
 The dominant poles of feedback are at -5 .
 The observer poles are 4 times faster than feedback poles.

$$s_{\text{observer}} = 4 \times -5 = -20.$$

Hence three poles are the same location.

$$s_1 = s_2 = s_3 = -20.$$

3) The faster pole is $s_3 = -3\zeta\omega_n = -3 \times 1 \times 5 = -15$

$$s_{\text{observer}} = 4 \times -15 = -60$$

Hence three poles are the same location

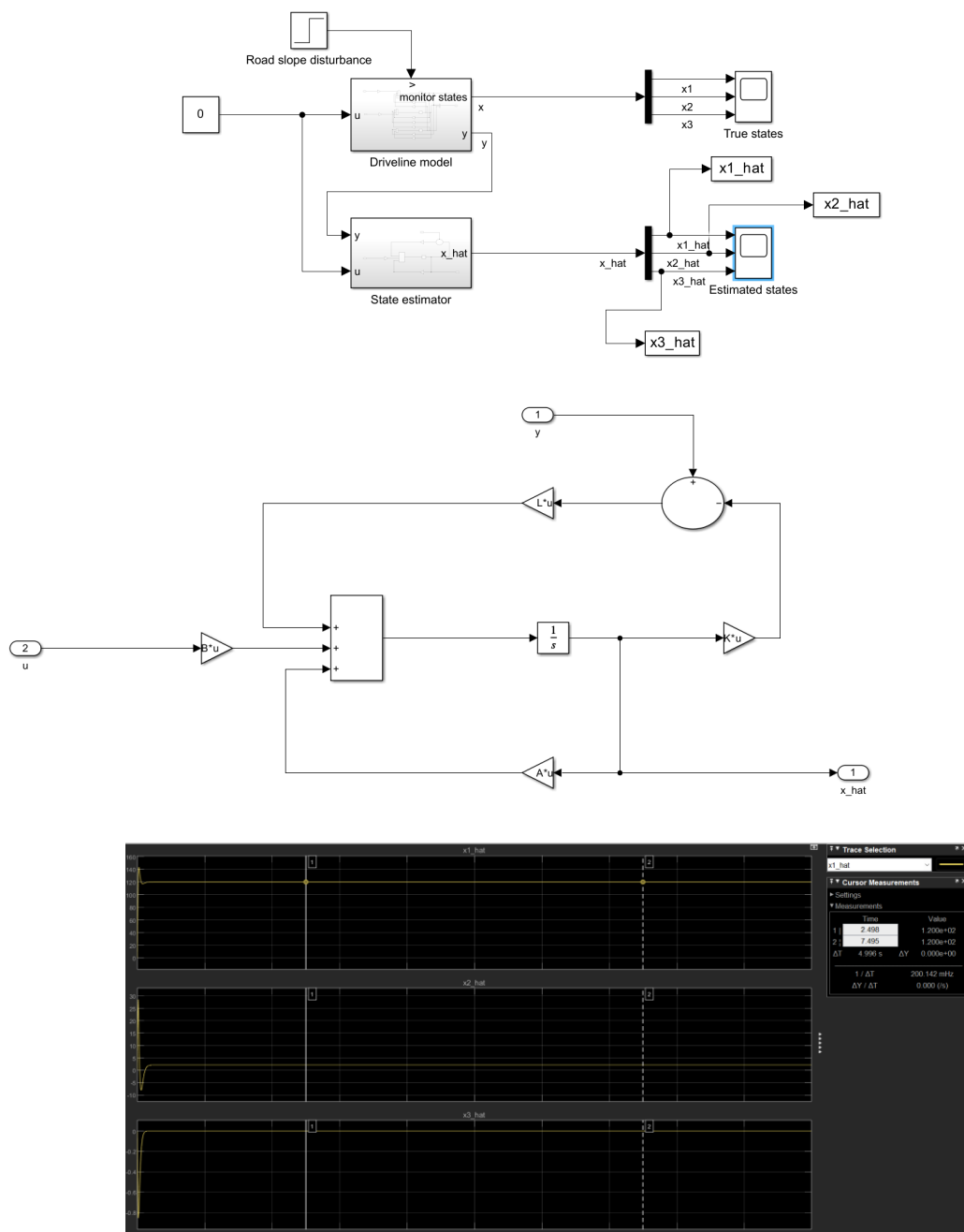
$$s_1 = s_2 = s_3 = -60.$$

The fastest pole is considered as the estimator poles = $[-60, -60, -60]$.

5. Determine the estimator gain, L , such that the estimator poles are placed according to Problem 4. Use the engine speed as measurement, $C = [1 \ 0 \ 0]$. To ease your design a little bit, it is suggested to use the design method presented in Theorem 8.2 in the textbook, a method very similar to Ackermann's formula for state feedback pole placement control design.
- The solution is provided in the MATLAB.
 The estimator gain matrix is $L_{pp} = [177.8610; 64.2256; -1.2881]$.
 Where $L = L_{pp}$.

6. Implement the observer! Add a screenshot of your implemented Simulink model in the written report! How fast will the estimated states converge to the true states, i.e., what is the settling time of the observer? As in the previous assignments, the settling time threshold is set to 5%. Study the system's transient response using the constant input in the Simulink model. As three states need to converge, we define settling time as the maximum settling times for the three states, i.e., the slowest of the states.
- The solution is provided in MATLAB and Simulink. (Refer 'A4_model_driveline_6')

Simulink model is



SettlingTime_6 = 0.2311; (The SettlingTime is considered of S2 which is the max value compare to the S1 and S3).

7. Design a Kalman filter for estimating the states of the driveline system1, with a road disturbance covariance of $R_v = 1!$ Answer with L! Remember the re-scaling of R_v as mentioned above.

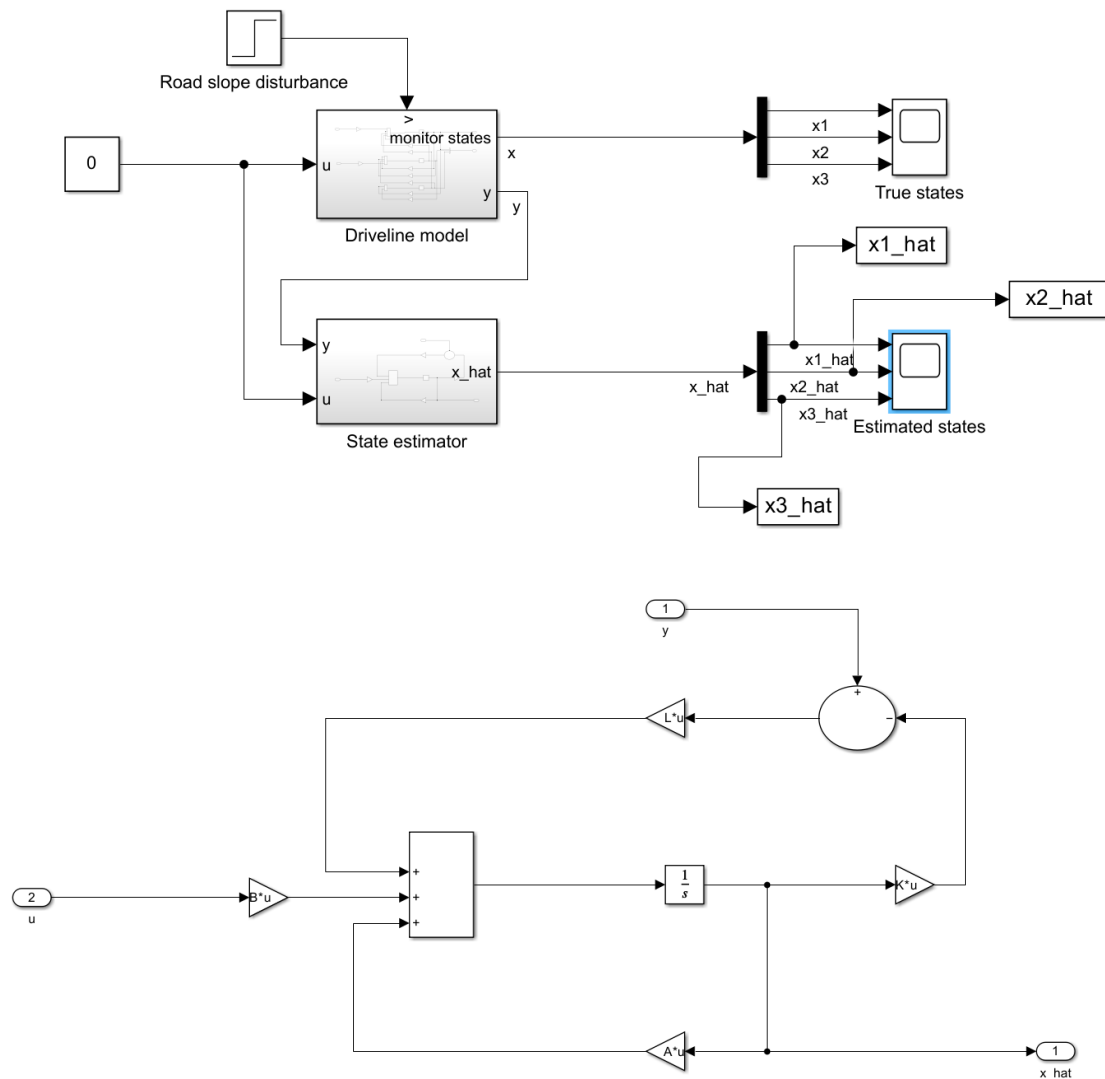
- The solution is provided in the MATLAB.

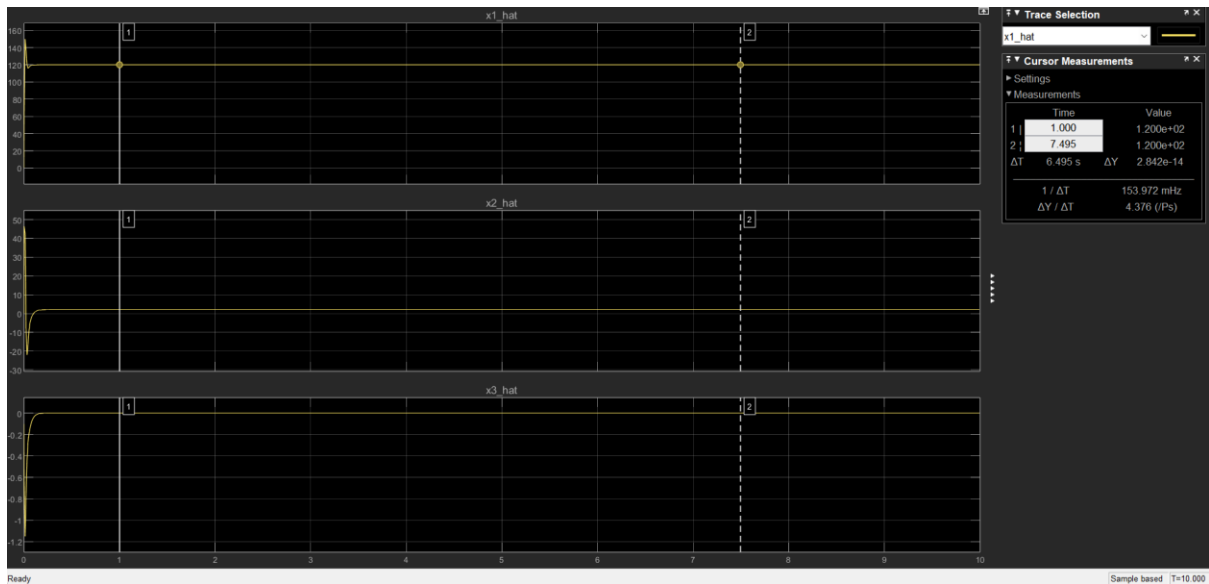
The Kalman filtering gain matrix = $L_kalman = [226.8342; 197.6810; -1.9560];$
Where $L = L_kalman$.

8. How fast will the estimated states converge to the true states for the Kalman filter, i.e., what is the settling time of the observer? Use the same threshold for settling time as in Problem 6.

- The solution is provided in MATLAB and Simulink.(Refer 'A4_model_driveline_8.slx')

Simulink model.





SettlingTime_8 = 0.2586. (The SettlingTime is considered of P2 which is the max value compare to the P1 and P3).

9. What value of the covariance R_v (unscaled) yields a similar performance as the pole placement controller? Try R_v equal to 0.01, 0.1, 1, 10 and 100?

- The solution is provided in MATLAB. (Refer 'A4_model_driveline_9.slx')

$R_v = 100$,

Simulink model

